

PRINTPI Production Setup Documentation

Machine Information

OLD PI (Production Monitoring / Print Server)

- Hostname: `printpi`
- Tailscale IP: `100.70.107.44`
- LAN IP: `10.72.9.183`
- OS: Debian 13 (Raspberry Pi)

NEW PI

- Hostname: `printpi-1`
- Tailscale IP: `100.108.118.38`

Installed Components

Core Services

SSH

Used for remote terminal access.

Check:

```
systemctl status ssh
```

Tailscale

Used for secure remote networking and remote SSH.

Check:

```
tailscale status
```

Dashboard:

- <https://login.tailscale.com/admin/machines>

CUPS (Print Server)

Used for printer management and print queue handling.

Check:

```
systemctl status cups  
lpstat -t
```

CUPS Dashboard:

```
http://10.72.9.183:631  
http://100.70.107.44:631
```

Watchdog

Used for automatic recovery during hangs/freezes.

Check:

```
systemctl status watchdog
```

UFW Firewall

Configured firewall.

Allowed Ports:

- 22 → SSH
- 631 → CUPS/IPP

Check:

```
sudo ufw status
```

Netdata Monitoring

Realtime monitoring dashboard.

Dashboards:

```
http://10.72.9.183:19999  
http://100.70.107.44:19999
```

Check:

```
systemctl status netdata
```

Printer Configuration

Active Printers

Check:

```
lpstat -t
```

Brother Queue

```
Brother_HL_L5210DN_series
```

Manual Test Print

```
lp -d Brother_HL_L5210DN_series /usr/share/cups/data/testprint
```

Queue Status

```
lpstat -o
```

Network Configuration

Ethernet Priority

- Ethernet (`eth0`) configured as primary network.
- WiFi (`wlan0`) configured as fallback hotspot.

Check:

```
ip route
```

Expected:

- Ethernet metric lower than WiFi metric.
-

Monitoring and Logging

Health Monitoring Script

File:

```
/home/printpi/healthcheck.sh
```

Purpose:

- Internet check
- Tailscale check
- CUPS check
- Printer queue check
- Undervoltage check

Manual Run:

```
~/healthcheck.sh
```

Log File:

```
/home/printpi/health.log
```

View Logs:

```
tail -f ~/health.log
```

Cron:

```
* /5 * * * * /home/printpi/healthcheck.sh
```

Alerts Script

File:

```
/home/printpi/alerts.sh
```

Purpose:

- Internet outage alerts
- CUPS failure alerts
- Queue stuck alerts
- Undervoltage alerts

Alert Email:

```
visionprintt@gmail.com
```

Manual Run:

```
~/alerts.sh
```

Cron:

```
* /5 * * * * /home/printpi/alerts.sh
```

CUPS Watchdog Script

File:

```
/home/printpi/cupswatch.sh
```

Purpose:

- Restart CUPS automatically if service fails.

Cron:

```
* * * * * /home/printpi/cupswatch.sh
```

Power and Stability Configuration

Sleep Disabled

Configured:

- sleep.target
- suspend.target
- hibernate.target
- hybrid-sleep.target

Purpose:

- Prevent kiosk sleeping.
-

Persistent Journal Logs

Enabled using:

```
sudo mkdir -p /var/log/journal
sudo systemctl restart systemd-journald
```

Purpose:

- Preserve logs across reboot.
-

Important Diagnostic Commands

Live System Logs

```
journalctl -f
```

Kernel / Hardware Logs

```
dmesg -w
```

Health Logs

```
tail -f ~/health.log
```

Failed Services

```
systemctl --failed
```

Printer Queue

```
lpstat -t
```

Tailscale

```
tailscale status
```

Check Undervoltage

```
vcgencmd get_throttled
```

Healthy:

```
throttled=0x0
```

Email Notifications

Configured using:

- msmtplib
- mailutils

Mail Sender:

```
hpsnowjug@gmail.com
```

Alert Receiver:

```
visionprintt@gmail.com
```

Mail Config:

```
~/ .msmtprc
```

Test Mail:

```
echo "Pi test mail" | mail -s "PRINTPI TEST" visionprintt@gmail.com
```

Current Operational Status

Production Features Enabled

- Remote SSH access
- Tailscale remote networking
- Ethernet/WiFi failover
- Watchdog recovery
- Persistent logging
- Health monitoring
- Email alerts
- Firewall protection
- Realtime monitoring dashboard
- Automatic CUPS recovery
- Sleep prevention

Known Risks / Future Improvements

Recommended Future Upgrades

UPS

Protect against:

- sudden power cuts
- filesystem corruption

SSD Boot

Improves:

- storage reliability
- write endurance
- long-term stability

Network Printer

More stable than USB printers.

FastAPI Print Service

For:

```
Backend → Pi API → CUPS → Printer
```

Telegram/Discord Alerts

Optional improvement over email alerts.

Important Paths

Scripts

```
/home/printpi/healthcheck.sh  
/home/printpi/alerts.sh  
/home/printpi/cupswatch.sh
```

Logs

```
/home/printpi/health.log  
~/msmtp.log
```

Configs

```
/etc/cups/cupsd.conf  
~/msmtp.rc
```

Last Notes

This Raspberry Pi is currently configured as a production-style remotely manageable print and monitoring server with:

- observability
- self-healing behavior
- remote management
- email alerting
- monitoring dashboard
- automatic recovery mechanisms